



The effect of environmental heterogeneity on the fitness of antibiotic resistance mutations in *Escherichia coli*

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Abstract

Antimicrobial resistance (AMR) is widespread, with resistant pathogens found not only in the clinic but throughout the natural and built environments. It is unclear, however, to what extent AMR should be expected to persist in the environment: Laboratory experiments have suggested that resistance is typically costly, and hence should be selected against. However, such studies may not generalize to broader environmental settings. As such, we estimated the fitness of six AMR mutants of *Escherichia coli* in ten media reflecting ecological niches, human hosts, and standard lab conditions. We found extensive evidence for genotype × environment interactions, with different genotypes favored in different environments. Resistance alleles did not suffer a cost in some environments, raising the possibility of environmental reservoirs of resistance. Interestingly, fitness tended to be higher in lower quality environments. Our findings emphasize the need to consider the ecology and evolution of resistance in the broader environment in the management of AMR.

Keywords Antibiotic resistance · Bacteria · Fitness · Genotype × environment

Introduction

Antimicrobial resistance (AMR) is a serious and growing threat to public health worldwide (O'Neill 2016). Amongst bacterial pathogens, resistance to many front-line antimicrobials is now widespread in the clinic—for example, about 26% of hospital isolates of *Escherichia coli* were resistant to quinolone antibiotics in a 2016 Canadian survey (Public Health Agency of Canada 2016). If AMR keeps evolving unchecked, it is predicted that deaths caused by AMR infections will increase from 700,000 per year to 10 million per year in 2050 (O'Neill 2014)—more than cancer, cholera, diabetes, measles, and diarrheal diseases combined (O'Neill 2016). There is thus increasing interest

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in understanding the mechanisms by which AMR pathogens persist and spread in the wild. This is fundamentally a question of the evolutionary ecology of resistance, since persistence and spread will be determined by interactions between AMR pathogens and the environment, other microbes, and host organisms.

Worryingly, resistant bacteria are found not only in clinical settings, but throughout the natural and built environments, including in wastewater, fresh water, and on terrestrial surfaces (Wichmann et al. 2014; Marti et al. 2014; Bengtsson-Palme et al. 2018). While it is known that many AMR genes are ancient (D'Costa et al. 2011; Perry et al. 2016), anthropogenic use of antimicrobials is thought to have increased the prevalence of AMR in the last century (Knapp et al. 2010). The presence of AMR in the environment is worrying for at least two reasons. First, there is concern that AMR infections may be acquired directly from environmental exposure. While the overall risk of AMR infection from the environment is unclear (Huijbers et al. 2015), there is epidemiological evidence linking recreational swimming with AMR infections (Søraas et al. 2013; Leonard et al. 2015; Schijven et al. 2015; Griffith et al. 2016).

Second, some environments could serve as reservoirs for AMR. Laboratory experiments have suggested that resistance mutations, as well as horizontally transferred resistance genes, are costly in the absence of antibiotics (Melnik et al. 2015; Vogwill and Maclean 2015; Carroll and Wong 2018). If this is true, then we would expect the prevalence of clinical resistance to decrease with reduced antibiotic use. Consistent with this expectation, drug restriction protocols in both human and animal populations are frequently successful (Tang et al. 2017). However, if a given resistance mutation or gene were selectively neutral or even beneficial in some natural environments, then we would expect it to be eliminated more slowly—or not at all—even following cessation of antibiotic use. Theory indicates that spatial structure can maintain polymorphism even if an allele is deleterious in some environments (Orr 2009). As such, if a mutation has high fitness in some environments, and/or spends prolonged periods of time in high fitness environments, it will be expected to persist.

Unfortunately, data are sparse concerning the fitness effects of resistance mutations in different environments (Lupo et al. 2012; Melnik et al. 2015). Typically, fitness is measured in a single laboratory medium, such as Lysogeny Broth (LB) or minimal medium. In the handful of cases where the fitness of an AMR mutation has been estimated in more than one environment, it is clear that fitness is strongly dependent on the assay environment (Remold and Lenski 2001; Hall et al. 2011; Hall 2013; Gifford et al. 2016; Hubbard et al. 2019). Moreover, when multiple resistance mutations are assayed in several environments, the rank order of fitness effects may vary, such that different mutants are most fit in different environments. For example, different rifampicin resistant mutants of *E. coli* varied in their ability to grow on different carbon sources (Hall 2013), with different rank orders of genotypes on different carbon sources. By contrast, Gifford et al. (2016) found that the rank order of fitness effects was consistent across several environmental treatments for four rifampicin resistant mutants of *Pseudomonas aeruginosa*.

Cases where the rank order of genotypes varies by environment are examples of genotype $\times$ environment ($G \times E$) interactions, a widespread evolutionary phenomenon whereby different genotypes respond differently to environmental variation (Via and Lande 1985; de Vos et al. 2018; Saltz et al. 2018; Burghardt 2019). $G \times E$ underlies local adaptation in experimental microcosms of bacteria (Kassen and Rainey 2004), and can maintain polymorphism in populations or meta-populations that occupy heterogeneous environments (Orr 2009). In the case of AMR, $G \times E$ has the potential to maintain resistance alleles in a

population, even if they are deleterious in some environments, since they may be neutral or beneficial in other environments.

Given the dearth of data on $G \times E$ for resistance mutations, we measured the fitness of 4 quinolone resistance mutations and 2 rifampicin resistance mutations across a range of environmental conditions in *E. coli*. Rifampicin disrupts transcription by binding to the β -subunit of RNA polymerase (RpoB), inhibiting transcript elongation and gene expression (Campbell et al. 2001). Resistance to rifampicin can evolve via mutations that alter the rifampicin binding pocket of RpoB, preventing the drug from binding properly (Campbell et al. 2001). Quinolones such as ciprofloxacin and nalidixic acid impair DNA strand separation, and by extension DNA replication, transcription, recombination, and repair, by binding to the α -subunit of DNA gyrase (GyrA) (Willmott and Maxwell 1993). Resistance to quinolones can occur by mutations in the *gyrA* gene that reduce binding of quinolones to the enzyme (Jacoby 2005). We used environments including media resembling human hosts, standard lab media, as well as natural environments. We found evidence for widespread $G \times E$, and at best modest correlations between environments. Our findings have important implications for the maintenance of resistance in natural populations.

Methods

All bacterial strains used in this study were derived from a standard laboratory strain of *E. coli*, K-12 (MG1655). A Lac⁻ derivative of MG1655, NCM520, was obtained from the *E. coli* Genetic Stock Center at Yale. Six AMR strains, each bearing a single chromosomal resistance mutation, were used. A quinolone resistant strain carrying a canonical *gyrA* S83L mutation was constructed by gene gorging (Herring 2003), as described in Bhatnagar et al. (2019). Additional spontaneous mutants resistant to quinolones (*gyrA* S83A; *gyrA* D87G; *gyrA* D87Y) or to rifampicin (*rpoB* I572L, *rpoB* I572S) were generated by plating independent overnight cultures of MG1655 on lysogeny broth (LB; 10 g/L tryptone, 5 g/L yeast extract, 10 g/L NaCl) supplemented with 1× or 2× MIC of the relevant antibiotic, as described in Wong and Seguin (2015). Resistance mutations were identified by whole genome sequencing of Nextera XT libraries on the Illumina MiSeq platform, using paired-end 300 bp reads, as described in Basra et al. (2018).

The competitive fitness of each resistant strain was estimated in ten environments using head-to-head competition assays. We chose media to reflect the diversity of environments in which AMR bacteria may be found, including media mimicking host environments (synthetic lower intestine (gluconate + M9), synthetic urine, synthetic colon), as well as media representing extra-host environments (river water, soil, sewage overflow). Primary and anaerobic sludge were obtained at the Robert O. Pickard Environmental Centre, Ottawa, Ontario. River water and soil were obtained from the Rideau river and its bank, by the HHJ Nesbitt Biology Building at Carleton University, Ottawa. Two common laboratory media, LB broth, and M9 minimal medium + glucose, were used because they are standard in microbiology experiments. In-depth description of these media preparations can be found in Table S1.

For the competition assays, cultures were streaked onto LB-agar plates and incubated at 37°, and 24 h later single colonies were inoculated into 2 mL of LB media and grown at 37 °C for 18 h in an orbital shaker (150 rpm). These populations were diluted 1:10 and acclimatized to each respective medium for 18 h in an orbital shaker (150 rpm). Then this culture was diluted 1:10 into the same medium in a 96-well plate for another 18 h in

an orbital shaker (150 rpm) at the corresponding temperatures (see Table S1) to ensure that the culture had acclimated to the media, and that any effects of the glycerol from the freezer stock were gone.

Each resistant genotype, along with the ancestral MG1655 strain, was then diluted 1:100 in the media of interest and mixed at a 1:1 ratio in 200 μ L of media in a 96-well plate, and the plates were allowed to mix in an orbital shaker (150 rpm) for 30 min. A 50 μ L aliquot of this initial T-0 h culture was frozen 1:1 with 50% glycerol at -80°C . The remaining 150 μ L of the sample was grown for 24 h in a 96-well plate in an orbital shaker (150 rpm) at the temperature corresponding to the medium (Table S1). After 24 h, the T-24 hr cultures were frozen 1:1 with 50% glycerol at -80°C . Competitive fitness assays were completed all at the same time to mitigate batch effects, such that all 6 AMR strains were competed against MG1655 in all 10 environments simultaneously. The assay was run using 4 biological replicates.

Mutant allele frequencies were estimated by next-generation sequencing (NGS). The protocol from Illumina's 16S Metagenomic Library Prep Guide (Illumina 2013) was adapted for this experiment. Region-specific primers, tagged with Illumina adapters, were used to amplify the locus containing the relevant AMR mutation. Primer sequences are given in Table S2. PCR amplicons from the T-0 h and T-24 h cultures were sequenced using an Illumina MiSeq, using paired-end 300 bp reads. Read quality was assessed with FastQC (Andrews 2010), and reads were trimmed with Trimmomatic (parameters: leading=20, trailing=20, window_length=4, window_qual=20, min_length=36) (Bolger et al. 2014). Quality was assessed again using FastQC and MultiQC was then used to assemble FastQC data for all files before and after trimming (Ewels et al. 2016). Paired reads were merged using Flash (parameters: min_overlap=20, max_overlap=250) (Magoč and Salzberg 2011). Finally, wild-type and mutant allele counts were obtained from the merged reads using "grep". Raw sequence data have been deposited in the NCBI short read archive (SRA accession PRJNA577095) and scripts can be found on Dryad (<https://doi.org/10.5061/dryad.zw3r22847>). For each mutant in each environment, the selection coefficient s was calculated as per Dykhuizen and Hartl (1983) using the allele counts.

$$s = \left(\ln \left(\frac{AMR \text{ at } T24}{AMR \text{ at } T0} \right) - \ln \left(\frac{WT \text{ at } T24}{WT \text{ at } T0} \right) \right) / \text{Number of generations}$$

Relative fitness (w) was then calculated as $1 + s$, where the units for both w and s are in per generation (Dykhuizen and Hartl 1983; Wong et al. 2012). Here, the number of generations is 6.66 ($\log_2(100)$).

All statistical analyses were performed in R version 3.5.1 (R Core Team 2018), and figures were prepared using GraphPad Prism (GraphPad Software 2014) or R. Analysis of variance was carried out via calls to the *lm* and *aov* functions in the R base package. Between-environment correlations (Fig. 2) were estimated using the *rcorr* function from the Hmisc package. Principal component analyses (Figure S2) were carried out using the *PCA* function in the FactoMineR package.

Yield assays were completed by inoculating 20 L of overnight culture (grown in each respective media in Table 2) into 180 μ L of the respective fresh media in a 96-well plate, with growth for 18 h in an orbital shaker at the temperatures indicated in Table S1. Plate counts were performed for 4 replicates per strain. Yield was measured in 6 out of the 10 environmental conditions due to shortages in the environmental media. Fresh collections of these media would differ in composition from the media used for the competitions (Table S1).

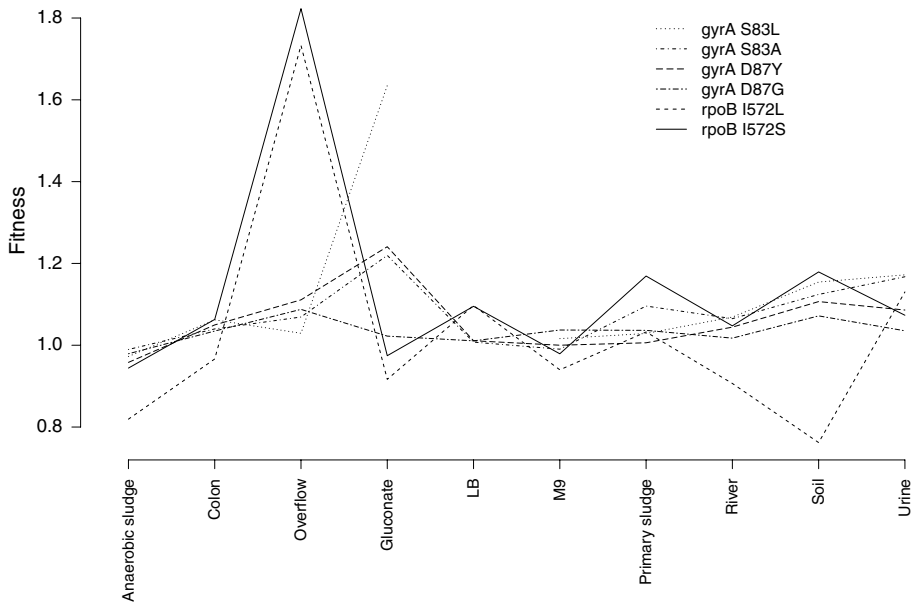


Fig. 1 Average relative fitness of AMR genotypes (legend) by environment. Head-to-head competitive fitness assays were performed for each AMR genotype versus the wildtype strain, MG1655, in the media of interest. Means of 4 replicate fitness measurements for each genotype in each environment are shown; error bars are omitted for clarity. Fitness above 1 indicates that a strain has a higher fitness than the wild type, and fitness below 1 indicates that a strain has lower fitness than the wild type. Fitness of *gyrA* S83L in LB is not included

Table 1 Two-way ANOVA for effects of genotype and environment on fitness for 6 AMR genotypes in 10 environments

Term	Df	Sum of squares	Mean square	F	P
Genotype	5	0.395	0.079	8.002	8.01×10^{-7}
Environment	9	2.233	0.248	25.116	$< 2 \times 10^{-16}$
Genotype $\times$ environment	44	4.522	0.103	10.405	$< 2 \times 10^{-16}$
Residuals	177	1.748	0.01		

Results

We sought to measure the impact of environmental variation on the fitness effects of AMR mutations. To do so, we estimated the fitness of six AMR mutants using head-to-head competitive fitness assays against an isogenic, drug susceptible ancestral strain. Fitness was estimated in 10 different media reflecting distinct niches, including host-like environments, exterior settings, and standard laboratory media. We found substantial and significant effects of genotype, environment, and their interaction on fitness (Fig. 1, Table 1).

The $G \times E$ interaction effect implies that the relative fitness effect of a given mutation is dependent on the assay environment. Notably, the rank order of fitness effects is not consistent across environments, as indicated by crossing reaction norms in Fig. 1; there is no

single genotype that is most fit across environments, and there is no single genotype that is least fit across environments. In a particularly striking example, the two *Rif^R* mutants (*rpoB* I572L and I572S) have markedly higher fitness than all other genotypes in sewage overflow. In other media, such as gluconate and anaerobic sludge, these same mutants are the lowest fitness genotypes. Similarly, *gyrA* D87Y and S83A have the highest fitness in gluconate, but intermediate fitness in most other media.

Given the strong $G \times E$ effect, it is difficult to interpret the main effects of environment and genotype. However, a few trends are apparent. Anaerobic sludge and M9 + glucose were the only two environments in which all assayed resistance mutations were neutral or costly such that the fitness values were approximately 1 or below (Figure S1). In all other environments, one or more resistance mutations were beneficial (Figure S1).

Although we found a significant interaction between genotype and environment (Table 1), this does not rule out the possibility that some environments closely resemble each other. As such, we asked to what extent fitness is correlated between environments (Fig. 2). Broadly, there were few strong correlations between media, and no single medium was representative of overall fitness. Nonetheless, fitness estimates were largely positively correlated between soil, artificial colon, anaerobic sludge, and M9 + glucose, with values of Pearson's ρ between 0.47 and 0.99. Similarly, in a principal component analysis (PCA; Figure S2), loadings are similar between these four environments.

We found a modest but significant correlation between fitness and yield, where we used yield as a proxy for environmental quality (Fig. 3; Table 2). Here, yield was measured as the population density, in CFU/mL, of the drug susceptible ancestral strain MG1655. Fitness was measured relative to the ancestral strain MG1655 in head-to head competition assays. Environments were rank-ordered according to yield, with higher ranks corresponding to lower yields; yield was highest in LB and lowest in soil medium. The positive correlation between rank and fitness thus indicates that mutant fitness is generally higher in lower yield environments.

Discussion

AMR pathogens, as well as resistance genes, are widespread in the environment (Finley et al. 2013; Wellington et al. 2013; Bengtsson-Palme et al. 2014; Berendonk et al. 2015). As bacterial community composition is shaped by an interplay of evolutionary, ecological and environmental factors, it is difficult to predict the fate of AMR organisms in the environment, or to obtain a clear understanding of the evolution and ecology of antibiotic resistance in this setting (Berendonk et al. 2015). The dissemination of AMR into the environment is concerning for at least two reasons. First, exposure to AMR pathogens in the environment may directly result in infections of humans or livestock. There is some evidence for environment-to-human transmission (Crombe et al. 2013; Wichmann et al. 2014; Huijbers et al. 2015) but more work is needed, particularly for pathogens with known or suspected environmental transmission pathways. Second, environmental niches in which resistance is selectively neutral or advantageous may act as reservoirs for AMR pathogens or genes (Berendonk et al. 2015; Huijbers et al. 2015). The presence of such cost-free niches may allow AMR to persist, even if resistance is costly in other environments.

How environmental heterogeneity affects the fitness of AMR bacteria is currently unclear, so it is difficult to evaluate the plausibility of environmental reservoirs. Most experimental estimates of the fitness effects of AMR mutations or plasmids are carried out

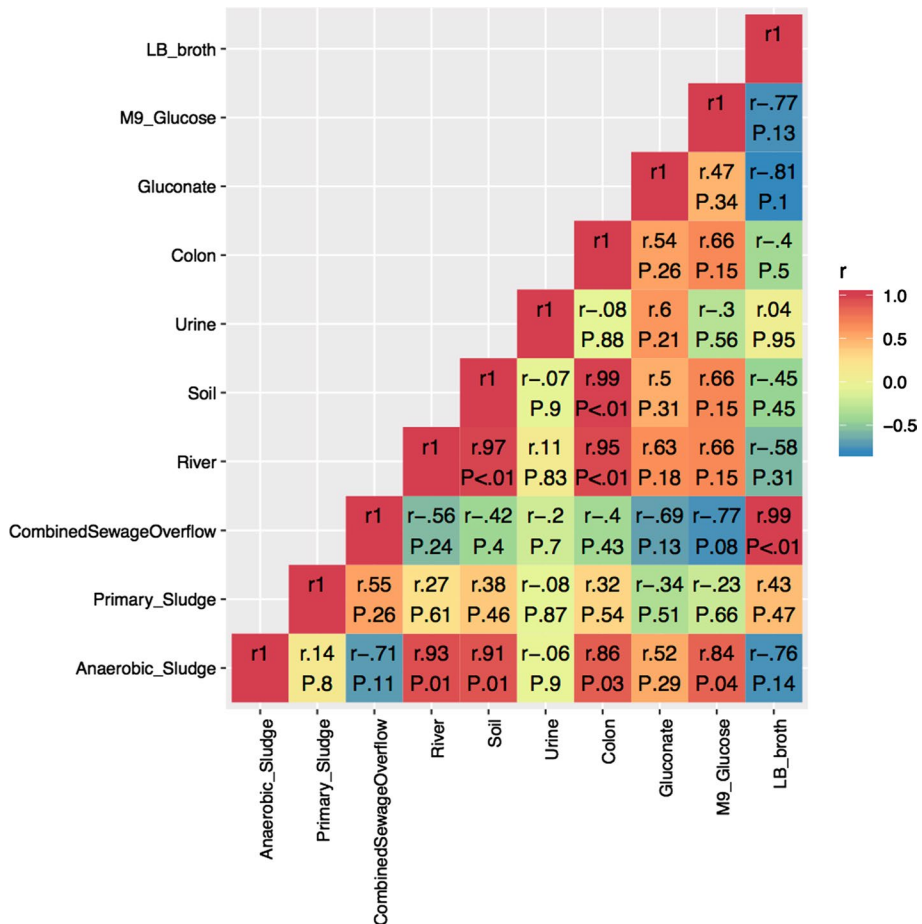


Fig. 2 Correlations in fitness between environments. Pearson correlation coefficients and significance levels were calculated from the mean fitness of six AMR in each environment. The colour coordinated legend (“r” for Pearson’s rho) indicates the value and sign of Pearson’s correlation coefficient, which is also indicated as a value on the plot (r)

in rich laboratory media; in such environments, resistance is often, but not always, costly (Melnik et al. 2015; Vogwill and Maclean 2015). The handful of studies that have investigated the effects of environmental variation have found that fitness can vary widely among environments, with mixed evidence for genotype $\times$ environment interactions—in this context, cases where the rank order of fitness effects of different mutants changes with environment (Remold and Lenski 2001; Bataillon et al. 2011; Hall 2013; Gifford et al. 2016; Sabarwal et al. 2016; Hubbard et al. 2019).

In an effort to connect laboratory fitness estimates to ecologically relevant environments, we investigated the fitness effects of six AMR mutations in *E. coli* in ten media reflecting laboratory settings, human infections, and broader environmental niches. We found widespread environmental variation in fitness effects, with substantial evidence for $G \times E$ interactions (Fig. 1; Table 1). Notably, all resistance mutations were neutral or beneficial in at least one environment; indeed, there were only two environments (M9 + glucose and

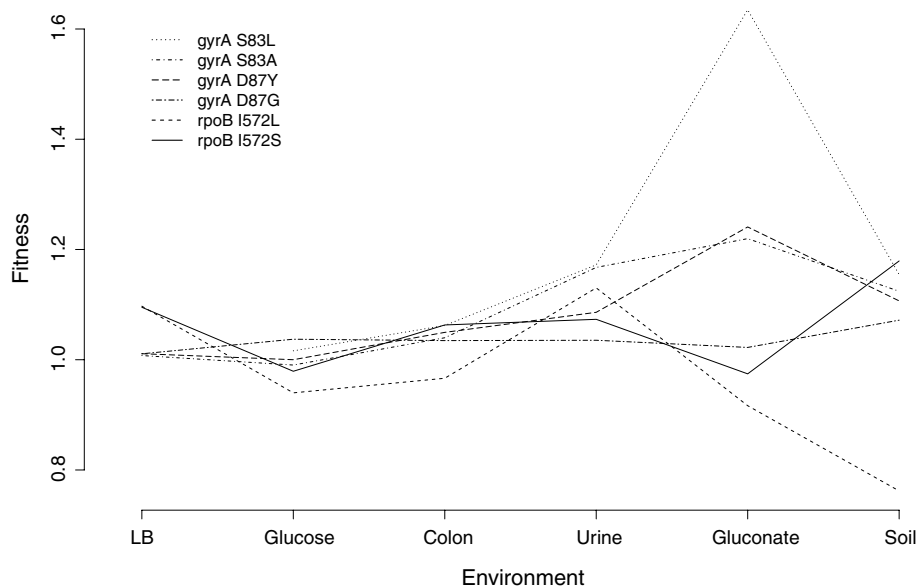


Fig. 3 Relationship between environmental productivity and fitness. Environments (x-axis) are rank-ordered according to the yield of the wild-type ancestral strain *E. coli* K-12 (MG1655) in that environment, as estimated from plate counts

Table 2 Linear model on the effects of yield on fitness. Environments were rank-ordered according to yield, with higher ranks indicating lower productivity

Term	Estimate	Std. error	t	P
Intercept	0.951685	0.063076	15.088	< 2e−16
Environment rank	0.039954	0.016196	2.467	0.015
Strain— <i>gyrA</i> D87Y	0.006911	0.089202	0.077	0.9384
Strain— <i>gyrA</i> D87G	0.057572	0.089202	0.645	0.5198
Strain— <i>gyrA</i> S83L	0.175192	0.089202	1.964	0.0517
Strain— <i>rpoB</i> I572L	0.067402	0.089202	0.756	0.4513
Strain— <i>rpoB</i> I572S	−0.083497	0.110642	−0.755	0.4518
<i>gyrA</i> D87Y × env	−0.004586	0.022905	−0.2	0.8416
<i>gyrA</i> D87G × env	−0.0325	0.022905	−1.419	0.1584
<i>gyrA</i> S83L × env	−0.085139	0.022905	−3.717	0.0003
<i>rpoB</i> I572L × env	−0.028062	0.022905	−1.225	0.2228
<i>rpoB</i> I572S × env	0.044984	0.026859	1.675	0.0964

Significant terms are indicated in bold

anaerobic sludge) in which all mutations were neutral or deleterious (Figure S1). While we recognize that our experiments do not capture all of the complexities of the natural world, our findings do suggest that some environmental niches could act as reservoirs for AMR.

From a practical perspective, it would be helpful to know which medium is most representative of the fitness effects of AMR mutations across environments. Here, correlations amongst environments were varied (Fig. 2), and largely modest. M9 minimal medium + glucose is perhaps the most representative environment, showing positive

correlations with soil, artificial colon, and anaerobic sludge. Notably, LB broth only shows a positive correlation with combined sewage overflow. LB broth is a standard medium for laboratory microbiology, and our findings suggest that it may not be representative of environments in the natural world. Future studies should estimate fitness in a broader selection of media to increase the generality of their findings. Furthermore, sampling of a greater range of resistance mutations and genes may help to resolve this issue.

It is uncertain what environmental factors determine the fitness effects of resistance mutations—or, more broadly, of any mutations. Temperature, nutrient availability, complexity, and other factors may all contribute to variation in fitness. Environmental stress has been proposed to be an important determinant of fitness effects (Agrawal and Whitlock 2010). Bacteria may encounter high stress environments quite frequently. For example, pathogens are faced with immune pressures and host nutrient restriction, and extra-intestinal environments are thought to be particularly stressful for *E. coli* (Savageau 1983). Environmental stress is sometimes thought to increase selection against deleterious mutations, but evidence for such a relationship is mixed (Agrawal and Whitlock 2010). Indeed, theory suggests that stress can have a variety of consequences on the fitness effects of new mutations, depending on how that stress affects the underlying fitness landscape (Martin and Lenormand 2006). Here, we found that competitive fitness tended to be higher in lower quality (higher stress) environments (Fig. 3, Table 2). This indicates that costs of resistance are lower—and that beneficial effects of resistance mutations are higher—in low yield environments. Interestingly, in at least one study in natural environments, multi-drug resistance was less prevalent in high carbon sediments than in low carbon sediments (McLain and Williams 2014), consistent with our findings. Findings in two recent studies on *Pseudomonas aeruginosa* (Lin et al. 2018) and *E. coli* (Basra et al. 2018) also found fewer costs of AMR in minimal media than in rich media. Whether our observations extend to other pathogens and environments—particularly environments that do not support sustained growth (Savageau 1983)—remains to be seen.

Resistance mutations are strongly beneficial in a specific environment (i.e., one containing antibiotic), and are hence unlikely to be representative of new mutations generally. It is unclear whether the patterns observed here—substantial effects of environment and of $G \times E$, and a modest relationship between environmental quality and fitness effects—will hold for other classes of mutations. It is interesting to note that, in mutation accumulation lines of *Arabidopsis thaliana*, environment had a stronger effect on phenotypic and fitness variation than did new mutations (Rutter et al. 2018). Similarly, in our study, environment and $G \times E$ have stronger effects on fitness than does genotype (Table 1); whether neutral and deleterious mutations show similar patterns in *E. coli* or other bacteria is not currently known.

That AMR is widespread in the environment is well-established. One challenge that we now face is to identify the factors determining the distribution of environmental AMR (Singer et al. 2016). Our findings indicate that environmental heterogeneity can have profound effects on the fitness effects of AMR mutations. This observation raises the possibility that some niches may act as environmental reservoirs for AMR. This underlines the importance of understanding the role of the environment on the evolution of resistance. An understanding of the mechanisms governing the interaction between AMR and environmental heterogeneity will help to predict—and ultimately intervene in—the distribution of AMR in the environment.

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